A HOW-TO GUIDE

Ask better questions of your energy data

Five questions anyone can ask an AI tool to turn smart meter readings into changes that lower the bill.

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WHY IT MATTERS

Your bill shows a total. Your meter records every half-hour.

48

readings a day from a smart meter, one every 30 minutes

1

number on your bill, with no sign of when or why you used it

30 min

on a demand tariff, one busy half-hour can set a charge for the whole month

BEFORE YOU START

You need three things

Your interval data

Ask your retailer or distributor for half-hourly readings as a CSV. Several months beats a few weeks.

Your tariff details

Your rates and any peak or demand window, from your plan or latest bill.

An AI tool

One that can read a spreadsheet you upload, such as Claude, Gemini or ChatGPT.

Strip personal details first. Delete your name, address, account and meter numbers. The AI only needs dates, times and kWh.

THE FRAMEWORK

Five questions, asked in order

Ask them in one conversation so each answer builds on the last. Treat the AI like an analyst you are briefing, not a search engine.

STEPS 1–4: FIND WHAT'S HAPPENING

STEP 5: DECIDE

1 Is the data healthy?

2 What is my baseload?

3 Where are the spikes?

4 Does anything look upside down?

5 What should I change first?

STEP 1 OF 5 · DATA HEALTH

Is the data healthy?

Why it matters

Meter exports are often messy: missing days, duplicate rows, labels that don't match the dates. Fix them first, or every later number is wrong.

THE PROMPT

"I have uploaded my half-hourly energy data. Act as a data analyst. Summarise the date formats, time intervals and units. Check for missing values, duplicate days, or columns that look wrong, such as day-of-week labels that don't match the dates. List what we need to fix before analysing further."

IN THE WORKED EXAMPLE · PREVIOUS HOME

108 of 274 dates, every 1st to 12th of the month, had the day and month swapped by the spreadsheet. Fixing that first meant every weekday comparison could be trusted.

STEP 2 OF 5 · BASELOAD

What is my baseload?

Why it matters

Baseload is what your home uses just by existing: fridge, router, devices on standby. It's the floor of your bill, and the place hidden waste shows up.

THE PROMPT

"Isolate the hours when my household is asleep, between [01:00 and 04:30]. What is my median consumption per half-hour in that window? What does it suggest about my always-on standby use, and is it high or low for a [one-bedroom apartment]?"

IN THE WORKED EXAMPLE · CURRENT HOME

The overnight floor was 0.064 kWh per half-hour. With everything off except the fridge, it fell to 0.037: about 40% of the floor was devices on standby.

STEP 3 OF 5 · SPIKES

Where are the spikes?

Why it matters

The biggest savings usually hide in a few large spikes, not in daily averages. Patterns tell you whether it's a habit, the season or a fault.

THE PROMPT

"Find the top 5 highest half-hour readings in the dataset. For each, give the date, time, day of week and kWh. Tell me whether they are one-off events or cluster into runs of consecutive days that suggest a seasonal or behavioural pattern."

IN THE WORKED EXAMPLE · PREVIOUS HOME

The five highest half-hours weren't scattered: all fell on 18 August 2025, the first day of a week of unusually high overnight use.

STEP 4 OF 5 · INVERSIONS

Does anything look upside down?

Why it matters

Comparing awake hours with sleeping hours catches things running when nobody needs them: heating, timers or a faulty appliance.

THE PROMPT

"Group my data into two blocks: evenings [18:00 to 00:00] and overnight [00:00 to 06:00]. Give the average for each. Then list the dates where overnight use was higher than evening use, and suggest what might explain each one."

IN THE WORKED EXAMPLE · PREVIOUS HOME

During that wet August week, overnight use ran at about 7× a normal night. The likely cause was the heating running through the night.

STEP 5 OF 5 · ACTION PLAN

What should I change first?

Why it matters

Data without a decision is just noise. This step turns findings into a short, ranked list, priced against your actual tariff.

THE PROMPT

"My plan is [your tariff, rates and any peak window]. Based on what you found, rank the top 3 changes that would lower my bill, such as moving appliance use to cheaper times. Estimate the saving for each and show your working."

IN THE WORKED EXAMPLE · CURRENT HOME

Top action: keep the dishwasher and dryer out of the 2–8 pm weekday window. One run there added $12.95 to $14.39 to the month; both together, $27.34.

BEFORE YOU ACT

Check the AI's work. It will make mistakes.

Recalculate one number

An AI summary said overnight use "surged by 700%". Its own figures, 11.74 vs 1.71 kWh, work out to about 7×: an increase of roughly 590%.

Test advice against your tariff

It also suggested pre-heating the home from 3 to 6 pm. On a demand tariff, that sits inside the peak window and would raise the bill.

Ask what's measured

It gave a fridge-only figure of 0.065 kWh as if measured. When the real data arrived, it was 0.037. Ask it to label every number.

MAKE IT STICK

Turn the findings into one page

Analysis doesn't change behaviour. A simple rule everyone in the house can follow does.

- Which appliances matter most
- Which times to avoid, and which are fine
- What one mistake costs, in dollars

Then put it on the fridge.



Household Energy Scheduling Cheat Sheet

Optimised for Red Energy Residential Demand Tariff

⚠️ **THE GOLDEN RULE: AVOID WEEKDAY AFTERNOONS (2 PM – 8 PM)**

Running heavy appliances between 2 PM and 8 PM Mon–Fri triggers a Demand Charge penalty of $0.2398 per kW/day ($13 to $27 extra on your monthly bill).

1. MASTER APPLIANCE SCHEDULE

Time Window	Days	Status	Action / Recommendation
14:00 – 20:00 (2 PM – 8 PM)	Monday – Friday	DANGER ZONE	**Do NOT run dishwasher or dryer.** High penalty risk.
07:00 – 14:00 (7 AM – 2 PM)	Monday – Friday	SAFE	Great for quick laundry washes & work-from-home runs.
20:00 – 07:00 (8 PM – 7 AM)	Monday – Friday	BEST	Ideal time to set timers for Dishwasher & Washing Machine.
All Day (07:00 – 22:00)	Saturday & Sunday	BEST	**Weekend Exempt!** Best time for Towel Wash + Dry cycles.

2. APPLIANCE-SPECIFIC GUIDELINES

🍽️ **Smeg Dishwasher (1800W)**

- **Mode:** ECO Mode preferred.
- **Best Time:** Run after 8:00 PM or set a delay timer overnight.
- **Avoid:** Never start between 2 PM & 8 PM weekdays.

🧺 **Fisher & Paykel Washer/Dryer (2000W)**

- **Quick / Everyday Wash:** Safe on weekday mornings or after 8 PM.
- **Towels (Wash + Dry):** Heavy 2000W load. **Run on Weekends only** or overnight.
- **Rule:** Never run dishwasher & dryer at the same time!

3. PEAK DEMAND PENALTY BREAKDOWN

What a single 30-minute run during 2 PM – 8 PM Weekdays adds to your bill:

Smeg Dishwasher (1.8 kW) alone during peak:	**+$12.95 / month penalty**
Fisher & Paykel Towel Wash + Dry (2.0 kW) alone during peak:	**+$14.39 / month penalty**
Both appliances running at the same time during peak:	**+$27.34 / month penalty**

Household Energy Guide • Red Energy Plan Reference

THE WORKED EXAMPLE · CURRENT HOME

What changed over the next three bills

<2 kW

peak demand in every bill since, down from 2.37 kW

~$5

a month less in demand charges from behaviour alone

~6 kWh

a day throughout: same energy, used at better times

Billing cycle (2026)	Peak demand	Demand charge	Cost per day
18 Jun – 17 Jul · before	2.37 kW	$17.05	$3.23
18 Jul – 17 Aug · first schedule	1.93 kW	$14.35	$3.15
18 Aug – 17 Sep · full schedule	1.69 / 1.57 kW	$9.74	$2.94

The last cycle is billed in two parts (August / September rates). The ~$5 compares all three cycles at the same winter rate.

YOUR TURN

Start this week

1

Download your interval data and strip your personal details

2

Ask the five questions in order, in one conversation

3

Check the maths, then put one page on the fridge

Guide by Matthew Edwards, change and transformation leader.

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